RECONSTRUCTION OF ABSORPTION COEFFICIENTS FOR QUANTITATIVE PHOTOACOUSTIC TOMOGRAPHY WITH UNSUPERVISED NEURAL REPRESENTATION

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ABSTRACT

Quantitative photoacoustic tomography (qPAT) combines the benefits of optical and ultrasound imaging. It reconstructs the quantitative distribution of optical parameters in biological tissues by detecting ultrasound signals generated from light absorption. The absorption coefficient (μ_a) is crucial as it is linked to physiological and pathological features. However, in qPAT imaging, due to wavelength-depend scattering and absorption, the photoacoustic image reflects the product of unknown light fluence (LF) and μ_a, rather than μ_a directly. Reconstructing μ_a from the photoacoustic image without knowing LF in deep tissue is an ill-posed inverse problem. To address this challenge, we propose a neural representation-based method for μ_a reconstruction. This method integrates a diffusion equation-based optical transport model, taking into account the optical processes in PAT imaging to enhance the accuracy and robustness of μ_a estimation. Simulated photoacoustic digital brain and phantom data experiments demonstrate that our proposed method can accurately reconstruct the μ_a distribution, thereby enhancing the quantitative imaging capabilities of qPAT.

Index Terms— Quantitative photoacoustic tomography, Absorption coefficient estimation, Unsupervised learning

1. INTRODUCTION

Photoacoustic imaging combines optical contrast with ultrasound resolution for deep-tissue visualization [1]. Quantitative PAT (qPAT) estimates absorption coefficients μ_a, but reconstruction is ill-posed due to unknown light fluence (LF) and tissue scattering [2]. Although FEM and Monte Carlo can model LF accurately, they are too slow for large heterogeneous tissues. Classical SBDC [3] and its segmentation-based variants [4] reduce LF influence but rely on manual tuning or strong assumptions, while iterative optimization [5] improves accuracy at high cost.

Recent deep learning methods [6, 7] enhance μ_a estimation but require large labeled datasets and ignore light transport physics. Moreover, all operate on discrete grids, increasing ill-posedness. Implicit Neural Representation (INR) offers a continuous, unsupervised alternative and has proven effective in sparse inverse problems [8, 9], yet remains underexplored for qPAT.

In this work, we propose ABCER, an unsupervised neural representation framework for directly reconstructing **AB**sorption **C**oefficient images from PAT images. As shown in Fig. 1, an MLP maps spatial coordinates to absorption values, guided by SBDC reconstruction as prior. Light fluence is modeled via FEM to enforce physical consistency, and the recovered μ_a enables adaptive meshing for finer structural resolution. Evaluations on digital brain and phantom datasets demonstrate superior accuracy over segmentation-based, optimization-based, and learning-based methods.

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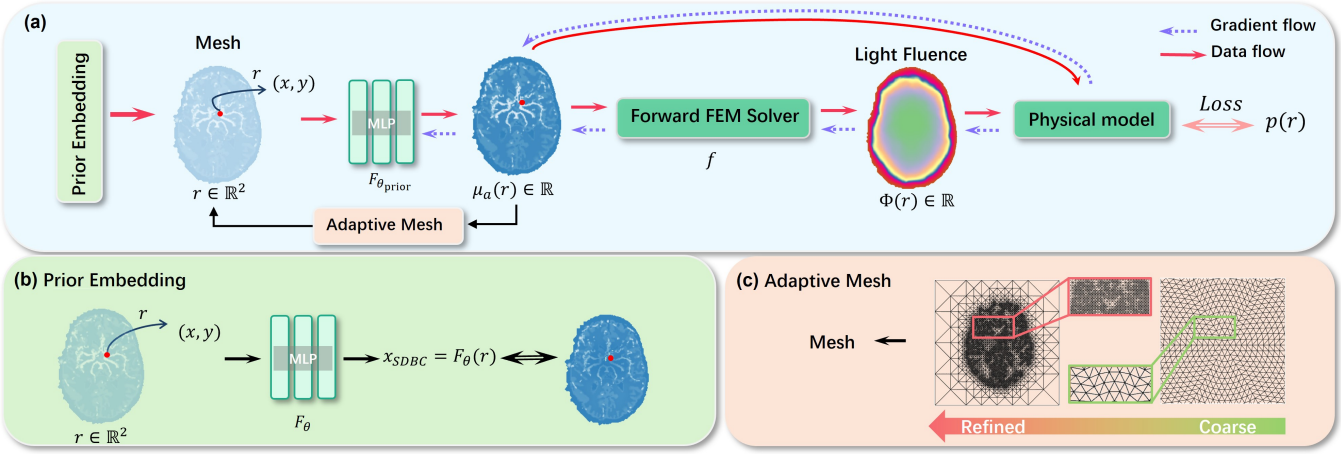


Fig. 1. Overview of ABCER framework. (a) An MLP continuously maps coordinates $r = (x, y)$ to μ_a, using X_{SBDC} as embedded prior and FEM-based fluence in the qPAT forward model for optimization. (b) X_{SBDC} provides structural guidance for stable convergence. (c) The recovered μ_a drives adaptive mesh refinement, increasing resolution in high-absorption regions.

2. METHODOLOGY

2.1. Problem Formulation

The qPAT process consists of two coupled components: light transport and acoustic signal generation. The initial pressure at location r is expressed as [5]:

$$p(r) = \Gamma \mu_a(r) \phi(\mu_a(r), \mu_s(r), g(r)), \tag{1}$$

where $\Gamma = \beta c^2 / C_p$ is the Grüneisen parameter (assumed constant [2]), $\mu_a(r)$ and $\mu_s(r)$ are absorption and scattering coefficients, ϕ is the deposited LF, and $g(r)$ is the anisotropy factor. Following [5], we assume the original PAT reconstruction is accurate and structural distortion is negligible. Thus the final pressure becomes:

$$p'(r) = \phi(\mu_a(r), \mu_s'(r)) \cdot \mu_a(r), \tag{2}$$

with the reduced scattering $\mu_s'(r) = \mu_s(r)(1 - g(r))$.

To model LF, the diffusion approximation yields:

$$-\nabla \cdot D(r) \nabla \phi(r, \omega) + \left(\mu_a(r) + \frac{i\omega}{c} \right) \phi(r, \omega) = q(r), \tag{3}$$

where $q(r)$ is the boundary light source and $D(r) = [3(\mu_a(r) + \mu_s'(r))]^{-1}$. We solve this equation using TOAST++ [10] with FEM-based discretization to obtain ϕ.

2.2. Absorption Coefficient Reconstruction via ABCER

Prior embedding. To stabilize optimization and guide structure recovery, the SBDC result is embedded as a prior [11], providing coarse anatomical cues and preventing convergence to degenerate solutions (Fig. 1b).

Data fidelity optimization. As shown in Fig. 1a, an MLP F_Θ represents $\mu_a(r)$ as a continuous function. Forward consistency is enforced by the FEM-based solver f, leading to the loss:

$$Loss = \| p(r) - F_\Theta(r) \cdot f(F_\Theta(r)) \|_2^2. \tag{4}$$

Coordinates are encoded via multiresolution hash encoding [12] ($L = 8, T = 2^{18}, F = 8, N_{\min} = 2, b = 2$), followed by a 3-layer MLP (64 ReLU neurons, Sigmoid output).

Adaptive meshing. To balance resolution and memory, we adopt iterative mesh refinement using `ameshref` [13]. Starting from a coarse grid, high-gradient absorption regions are selectively refined, enabling efficient ABCER training with fine structural recovery.

3. EXPERIMENTS

3.1. Datasets

Simulated photoacoustic digital brain. The dataset contains six tissue types (scattered scalp and skull re-

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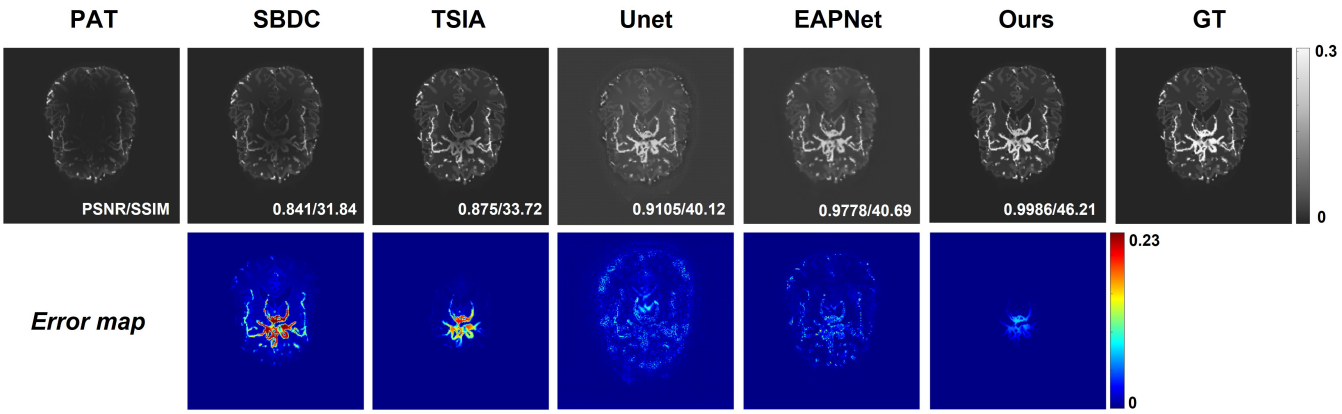


Fig. 2. Qualitative results of μ_a images by compared methods on photoacoustic digital brain.

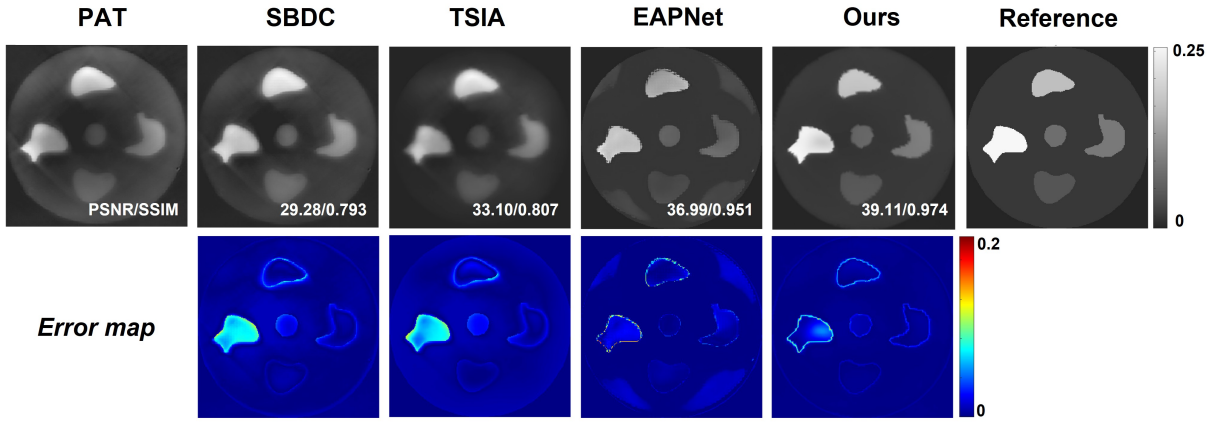


Fig. 3. Qualitative results of μ_a images by compared methods on phantom data.

moved to reduce scattering ambiguity), with optical parameters (μ_a, μ_s', g, n) from [14]. Following the setup in Section 3.1, 180 PAT images (256×256) were generated. For supervised baselines, 120 samples were used for training and 60 for testing.

Phantom experiment. To evaluate real-world performance, we used the public MSOT phantom dataset from [7]. It includes a 9.6-mm-radius phantom with five inclusions. Ground-truth μ_a was calibrated by preparing optical samples of known compositions, measuring transmittance/reflectance (LAMBDA 950, PerkinElmer), and recovering optical constants using the inverse adding-doubling (IAD) algorithm with anisotropy $g = 0.9$.

3.2. Baselines and Metrics

We compare ABCER against three representative reconstruction strategies: (1) SBDC (LF division), (2) TSIA (iterative optimization), and (3) supervised CNNs, including U-Net [15] and EAPNet [7]. Quantitative evaluation is conducted using PSNR and SSIM.

3.3. Results

Simulated brain. Table I shows that ABCER achieves the highest PSNR and SSIM. Fig. 2 confirms superior deep-tissue recovery. SBDC captures overall structure but fails in deeper regions, while TSIA improves accuracy yet lacks precision. U-Net and EAPNet partially recover structures but exhibit global inconsistencies.

Phantom study. Fig. 3 demonstrates that ABCER accurately reconstructs multiple inclusions with sharp boundaries. EAPNet produces reasonable contrast but exhibits edge errors in residual maps. SBDC and TSIA perform similarly due to limited tissue diversity. Table I further confirms ABCER's quantitative advantage.

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Table 1. Quantitative Comparison with other methods. We report the results over the photoacoustic digital brain data and phantom data.

Dataset	Method	SSIM↑	PSNR↑
Stimulated Data	SBDC [3]	0.8370	30.84
	TSIA [5]	0.8709	33.49
	Unet [15]	0.9105	40.12
	EAPNet [7]	0.9578	40.69
	Ours (ABCER)	**0.9786**	**46.21**
Phantom Data	SBDC [3]	0.7923	29.28
	TSIA [5]	0.8074	33.10
	EAPNet [7]	0.9518	36.99
	Ours (ABCER)	**0.9748**	**39.11**

Table 2. Quantitative results of our ABCER ablating the MLP network on the photoacoustic digital brain data.

Network	SSIM↑	PSNR↑
Fourier encoding [16]	0.9865	46.17
Hash encoding [12]	**0.9986**	**46.21**

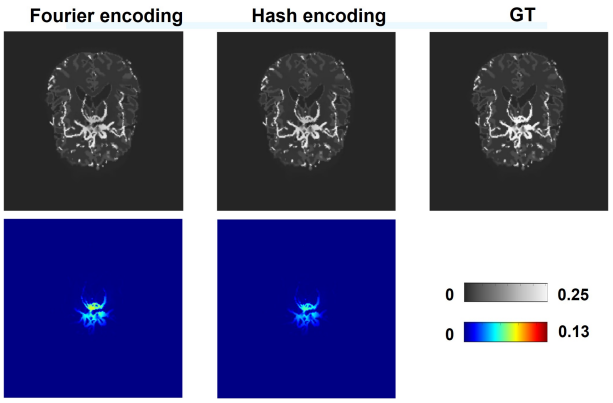


Fig. 4. Qualitative comparison of architectural variants on the digital brain dataset.

3.4. Ablation Study

To assess network sensitivity, we replace the hash encoder with Fourier encoding [16] and expand the MLP to six layers. As shown in Table II and Fig. 4, hash encoding yields slightly better interior accuracy. Overall, ABCER is robust to architectural variations, though advanced encoders may further accelerate convergence.

4. CONCLUSION

This paper presents ABCER, an unsupervised deep learning model for accurately recovering absorption coefficient distributions. ABCER uses SBDC results as prior information and combines an FEM-based physical model to precisely simulate photon scattering. An adaptive meshing strategy is introduced to cut resource consumption and boost computational efficiency. Verified by photoacoustic digital brain and calibrated phantom experiments, ABCER accurately recovers tissue absorption coefficients. Compared with classical segmentation-based, iteration-based, and other deep learning methods, ABCER shows significant performance improvement, validating the effectiveness of this unsupervised deep learning approach in the field.

5. COMPLIANCE WITH ETHICAL STANDARDS

This study was performed in line with the principles of the Declaration of Helsinki. Approval was granted by the Ethics Committee of local institution.

6. ACKNOWLEDGMENTS

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